

MTE 203 Midterm Exam General Information

MTE 203 – Advanced Calculus

June, 2026



UNIVERSITY OF
WATERLOO

MTE 203 Midterm Exam

The MTE 203 Midterm Exam is a two-hour exam

- Date: Monday June 22, 2026
- Time: 9:30am – 11:30am
- Location: E7 3343(Assigned Seats)*
E7 3353 (Assigned Seats)*

* Check your Odyssey account before the exam.

Integrity is a virtue in a successful professional career



ALLOWED

- UW ID
- Mathematical Handbook of Formulas and Tables, M. R. Spiegel (**no annotations allowed**)
- Scientific calculator approved by the UW Faculty of Mathematics or equivalent
- Pencil, eraser, sharpener
- Printed booklet with answer space will be provided
 - Formula Sheet #1 will be attached at the back

NOT ALLOWED

- Assignments
- Class notes or any other papers
- Telecommunication devices (e.g. cell phones, tablets, smart watches, any other wireless device, etc.)
 - You will be asked to put all your belongings, **including not approved calculators**, all these devices, by the side walls.

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Topics that will be covered

- **Vectors and 3D Analytical Geometry (Trim Chapter 11)**
 - Planes and lines (Trim 11.5)
 - Distance between points lines and planes (Trim 11.6)
 - Curves and surfaces (Trim 11.2)
 - Cylinders and quadric surfaces
 - Differentiation and integration of vectors (Trim 11.9)
 - Parametric and vector representation of curves (Trim 11.10)
 - Tangent vectors and length of curves (Trim 11.11)
 - Normal vectors, curvature and radius of curvature (Trim 11.12)
 - Displacement, velocity, and acceleration (Trim 11.13)
 - Normal and tangential components of acceleration

Topics that will be covered

- Differential Calculus of Multivariable Functions (Trim Chapter 12)
 - Multivariable functions (Trim 12.1)
 - Surfaces, level curves, and contour plots
 - Partial derivatives (Trim 12.3)
 - Higher order partial derivatives (Trim 12.5)
 - Laplace's equation, Wave Equation, etc.
 - Chain rules for partial derivatives (Trim 12.6)
 - Directional derivative, gradients (Trim 12.8, 12.4)
 - Tangent lines and tangent planes (Trim 12.9)

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Thank you

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